

Algebra II with Trigonometry

Chapter 7.2

Olympiad Solutions (Gemini)

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[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Solutions

1. Evaluate exactly: $\cos(\tan^{-1}(2) + \tan^{-1}(3))$.

Let $\alpha = \tan^{-1}(2)$ and $\beta = \tan^{-1}(3)$. Because $2 > 0$ and $3 > 0$, both α and β lie in the interval $(0, \frac{\pi}{2})$, meaning $\alpha + \beta \in (0, \pi)$.

Using the tangent addition formula, we find:

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{2 + 3}{1 - (2)(3)} = \frac{5}{-5} = -1.$$

Since $\alpha + \beta \in (0, \pi)$ and $\tan(\alpha + \beta) = -1$, it necessarily follows that $\alpha + \beta = \frac{3\pi}{4}$.

Thus, the exact value is:

$$\cos(\tan^{-1}(2) + \tan^{-1}(3)) = \cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$

2. Find the exact value of $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$.

Recall the tangent addition formula: $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$. Setting $A = 20^\circ$ and $B = 40^\circ$, we have:

$$\tan(60^\circ) = \frac{\tan 20^\circ + \tan 40^\circ}{1 - \tan 20^\circ \tan 40^\circ}.$$

Since $\tan 60^\circ = \sqrt{3}$, we can multiply both sides by the denominator:

$$\sqrt{3} - \sqrt{3} \tan 20^\circ \tan 40^\circ = \tan 20^\circ + \tan 40^\circ.$$

Rearranging the terms yields the given expression directly:

$$\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ = \sqrt{3}.$$

3. Compute the exact value of $\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ}$.

Combine the terms over a common denominator:

$$\frac{\cos 10^\circ - \sqrt{3} \sin 10^\circ}{\sin 10^\circ \cos 10^\circ}$$

To reveal a hidden angle addition, multiply the numerator and denominator by 2:

$$\frac{2 \left(\frac{1}{2} \cos 10^\circ - \frac{\sqrt{3}}{2} \sin 10^\circ \right)}{\frac{1}{2} (2 \sin 10^\circ \cos 10^\circ)}$$

Recognizing that $\sin 30^\circ = \frac{1}{2}$ and $\cos 30^\circ = \frac{\sqrt{3}}{2}$, the numerator simplifies using the sine difference formula, while the denominator simplifies using the sine double-angle formula:

$$\frac{2(\sin 30^\circ \cos 10^\circ - \cos 30^\circ \sin 10^\circ)}{\frac{1}{2} \sin 20^\circ} = \frac{4 \sin(30^\circ - 10^\circ)}{\sin 20^\circ}$$

This perfectly collapses to:

$$\frac{4 \sin 20^\circ}{\sin 20^\circ} = 4.$$

4. Find the exact value of $\sec^2 \left(\frac{1}{2} \sin^{-1} \left(\frac{4}{5} \right) \right) + \csc^2 \left(\frac{1}{2} \cos^{-1} \left(\frac{3}{5} \right) \right)$.

Let $\theta = \sin^{-1} \left(\frac{4}{5} \right)$. Because $\frac{4}{5} > 0$, θ is in the first quadrant. By the Pythagorean identity, $\cos \theta = \sqrt{1 - \left(\frac{4}{5} \right)^2} = \frac{3}{5}$. Consequently, $\cos^{-1} \left(\frac{3}{5} \right)$ is exactly the same angle θ .

The expression then beautifully transforms into a symmetric form:

$$\sec^2 \left(\frac{\theta}{2} \right) + \csc^2 \left(\frac{\theta}{2} \right)$$

Converting to sine and cosine, we get:

$$\frac{1}{\cos^2(\theta/2)} + \frac{1}{\sin^2(\theta/2)} = \frac{\sin^2(\theta/2) + \cos^2(\theta/2)}{\sin^2(\theta/2) \cos^2(\theta/2)}$$

Using the Pythagorean identity in the numerator and the double-angle formula $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$ in the denominator, this becomes:

$$\frac{1}{\frac{1}{4} \sin^2 \theta} = \frac{4}{\sin^2 \theta}$$

Since $\sin \theta = \frac{4}{5}$, the exact value is:

$$\frac{4}{(4/5)^2} = \frac{4}{16/25} = \frac{25}{4}.$$

5. Evaluate exactly: $\sin^2 10^\circ + \sin^2 50^\circ + \sin^2 70^\circ$.

Apply the power-reduction formula, $\sin^2 x = \frac{1 - \cos 2x}{2}$, to each term:

$$\frac{1 - \cos 20^\circ}{2} + \frac{1 - \cos 100^\circ}{2} + \frac{1 - \cos 140^\circ}{2}$$

This separates into a constant part and a sum of cosines:

$$\frac{3}{2} - \frac{1}{2}(\cos 20^\circ + \cos 100^\circ + \cos 140^\circ)$$

We can evaluate the sum of the last two cosines using the sum-to-product formula $\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$:

$$\cos 100^\circ + \cos 140^\circ = 2 \cos(120^\circ) \cos(20^\circ) = 2 \left(-\frac{1}{2} \right) \cos 20^\circ = -\cos 20^\circ$$

Substituting this back into our expression, the cosine terms cancel entirely:

$$\frac{3}{2} - \frac{1}{2}(\cos 20^\circ - \cos 20^\circ) = \frac{3}{2}.$$

6. Compute the exact integer value of the product:

$$\prod_{k=1}^{59} (1 + \sqrt{3} \tan k^\circ).$$

First, rewrite the typical term inside the product using sine and cosine:

$$1 + \sqrt{3} \tan k^\circ = 1 + \frac{\sqrt{3} \sin k^\circ}{\cos k^\circ} = \frac{\cos k^\circ + \sqrt{3} \sin k^\circ}{\cos k^\circ}$$

Factor out a 2 in the numerator to construct an angle addition identity:

$$\frac{2 \left(\frac{1}{2} \cos k^\circ + \frac{\sqrt{3}}{2} \sin k^\circ \right)}{\cos k^\circ} = \frac{2(\sin 30^\circ \cos k^\circ + \cos 30^\circ \sin k^\circ)}{\cos k^\circ} = \frac{2 \sin(k^\circ + 30^\circ)}{\cos k^\circ}$$

Now, substitute this back into the product:

$$\prod_{k=1}^{59} \frac{2 \sin(k^\circ + 30^\circ)}{\cos k^\circ} = 2^{59} \frac{\prod_{k=1}^{59} \sin(k^\circ + 30^\circ)}{\prod_{k=1}^{59} \cos k^\circ}$$

Observe the structure of the numerator and the denominator:

- The numerator expands to: $\sin 31^\circ \sin 32^\circ \cdots \sin 89^\circ$.
- The denominator expands to: $\cos 1^\circ \cos 2^\circ \cdots \cos 59^\circ$.

By the complementary angle identity, $\cos \theta = \sin(90^\circ - \theta)$. Therefore, the sequence of terms in the denominator can be rewritten identically as:

$$\sin 89^\circ \sin 88^\circ \cdots \sin 31^\circ$$

The numerator and the denominator contain exactly the same factors, simply multiplied in reverse order. They cancel completely, leaving the exact integer value:

$$2^{59}.$$